A UNIQUE-POINT STACK CONTAINS A UNIQUE STRICTLY FULL REDUCED LOCALLY NOETHERIAN ONE-POINT SUBSTACK

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2790-2802 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 2804-2856, the complete author proof (51 body lines).

Editorial note. Navigation stubs: 11 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 3 of queue_stacks-properties.json (pb=2019, stmt 2790-2802, proof 2804-2856).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04Y5: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2856–2856. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03YQ: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2856–2856. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04T4: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2856–2856. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06FH: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2856–2856. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06FI: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2856–2856. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-monomorphism: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1674–1687. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-monomorphism-into-point: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2759–2764. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-points-cartesian: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1019–1036. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-unique-point: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2654–2662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-unique-point-better: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2710–2717. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06G4: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2856–2856. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{Z}$ be an algebraic stack. Assume $\mathcal{Z}$ satisfies the equivalent conditions of Lemma lemma-unique-point. Then there exists a unique strictly full subcategory $\mathcal{Z}' \subset \mathcal{Z}$ such that $\mathcal{Z}'$ is an algebraic stack which satisfies the equivalent conditions of Lemma lemma-unique-point-better. The inclusion morphism $\mathcal{Z}' \to \mathcal{Z}$ is a monomorphism of algebraic stacks.*

Proof. The last part is immediate from the first part and Lemma lemma-monomorphism. Pick a field k and a morphism $\mathrm{Spec}(k) \to \mathcal{Z}$ which is surjective, flat, and locally of finite type. Set $U = \mathrm{Spec}(k)$ and $R = U \times_{\mathcal{Z}} U$. The projections $s, t : R \to U$ are locally of finite type. Since U is the spectrum of a field, it follows that s, t are flat and locally of finite presentation (by Morphisms of Spaces, Lemma 06G4). We see that $\mathcal{Z}' = [U/R]$ is an algebraic stack by Criteria for Representability, Theorem 06FI. By Algebraic Stacks, Lemma 04T4 we obtain a canonical morphism

$$f : \mathcal{Z}' \longrightarrow \mathcal{Z}$$

which is fully faithful. Hence this morphism is representable by algebraic spaces, see Algebraic Stacks, Lemma 04Y5 and a monomorphism, see Lemma lemma-monomorphism. By Criteria for Representability, Lemma 06FH the morphism $U \to \mathcal{Z}'$ is surjective, flat, and locally of finite presentation. Hence $\mathcal{Z}'$ is an algebraic stack which satisfies the equivalent conditions of Lemma lemma-unique-point-better. By Algebraic Stacks, Lemma 03YQ we may replace $\mathcal{Z}'$ by its essential image in $\mathcal{Z}$. Hence we have proved all the assertions of the lemma except for the uniqueness of $\mathcal{Z}' \subset \mathcal{Z}$. Suppose that $\mathcal{Z}'' \subset \mathcal{Z}$ is a second such algebraic stack. Then the projections

$$\mathcal{Z}' \longleftarrow \mathcal{Z}' \times_{\mathcal{Z}} \mathcal{Z}'' \longrightarrow \mathcal{Z}''$$

are monomorphisms. The algebraic stack in the middle is nonempty by Lemma lemma-points-cartesian. Hence the two projections are isomorphisms by Lemma lemma-monomorphism-into-point and we win. □